

Answer Key — Spherical Mirrors

Final answer with a one-line reason. Full methods are in the Notes' worked examples.

Objective

- Q1 **(b) concave** — an inward-curving mirror is concave.
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- Q2 **(b) erect and diminished** — convex mirrors give erect, diminished images.
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- Q3 **(c) concave mirror** — concave mirrors are used in torch/headlight reflectors.
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- Q4 **(b) give a wide, erect view** — convex mirrors give a wide erect view to prevent collisions.
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- Q5 **(a) Both true, R explains A** — the small image makes objects seem far, so the warning is needed — as R explains.
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- Q6 **concave** — a concave mirror (dentist's mirror).
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- Q7 **same** — a plane mirror gives a same-size erect image.
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Short answer

- Q8 Concave curves inward and can give an inverted (or enlarged, when close) image; convex curves outward and always gives an erect, diminished image.
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- Q9 Concave: torch/headlight reflector, dentist's mirror, solar concentrator (any two). Convex: vehicle side mirror, road-safety/surveillance mirror (any two).
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- Q10 A plane mirror gives an erect, same-size image always; in spherical mirrors the size changes with distance, and a concave mirror can also invert the image.
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Long / application

- Q11 The enlarged (and possibly inverted when farther) image is from a **concave** mirror; close up it gives an enlarged erect image. The tiny erect image is from a **convex** mirror, which always forms a diminished, erect image. The curvature direction (inward vs outward) causes the difference.
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- Q12 (i) Concave — it concentrates light into a strong beam. (ii) Convex — erect, diminished image gives a wide view to monitor a large area. (iii) Concave — held close, it gives an enlarged erect image of the face.
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